NATIONAL POLYTECHNIC INSTITUTE OF CAMBODIA

FACULTY OF ELECTRONIC

DEPARTMENT OF ELECTRONIC

Assignment

Power Electronic 2

120W Buck Converter Design

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1 Overview Of Power Electronic

1.1 What's DC/DC Converter?

A DC-DC converter is an electronic circuit that converts a source of direct current (DC) from one voltage level to another. Unlike transformers, which are designed to work with alternating current (AC), DC-DC converters are specifically tailored to handle DC power. The primary purpose of a DC-DC converter is to regulate and transform voltage levels to meet the requirements of various electronic devices. For example, step-down converters reduce voltage (for example, from 12V to 5V), while step-up converters increase voltage (e.g., from 3V to 12V).
There are several types of DC-DC converters, including:

- **Buck Converter** : Steps down voltage.
- **Boost Converter** : Steps up voltage.
- **Buck-Boost Converter** : Can both step up and step down voltage.
- **Flyback Converter** : Used for isolation and multiple outputs.

DC-DC converters are used in a wide range of applications, including consumer electronics (smartphones, laptops, and tablets), automotive systems (electric vehicles, infotainment systems, and lighting), renewable energy (solar panels, battery storage systems, and wind turbines) and industrial equipment (motor drives, robotics, and control systems).

1.2 Importance of Buck Converters

A buck converter is a type of DC-DC converter that steps down voltage while maintaining high efficiency. It is also known as a step-down converter because its output voltage is always lower than the input voltage. Buck converters are critical in modern electronics because they allow devices to operate at lower voltages while drawing power from higher-voltage sources. For example, a smartphone battery operates at around 3.7V, but the charger provides 5V or more. A buck converter steps this down efficiently.
The key advantages of buck converters include the following:

- **High Efficiency** : Buck converters typically achieve efficiencies of 90 or higher, minimizing energy loss.
- **Compact Size** : Modern buck converters are small and lightweight, making them ideal for portable devices.
- **Cost-Effective** : They use fewer components compared to other types of converters, reducing cost and complexity.
- **Reliability** : With proper design, buck converters can operate reliably for years in demanding environments.

Buck converters are widely used across industries:

- **Consumer Electronics** : Power microprocessors, memory chips, and displays in laptops, smartphones, and gaming consoles.
- **Automotive Systems** : Regulating voltage for LED headlights, information system and electric vehicle drivetrains.

- **Renewable Energy** : Converting the solar panel output to usable voltages for battery charging and grid integration.
- **Telecommunications** : Providing stable power for routers, switches, and communication towers.

2 Fundamental Of Buck Converter

A buck converter is a type of DC-DC switching converter that steps down a higher input voltage V_{in} to a lower output voltage V_{out}. Below is a detailed, step-by-step breakdown of its operation, components, and key relationships.

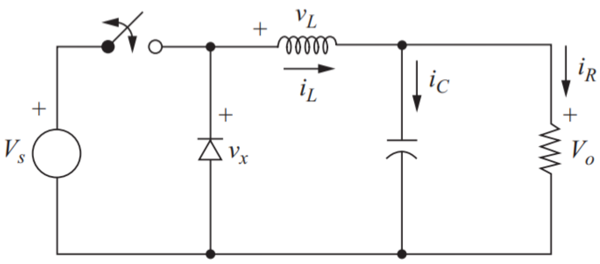

Figure 1: *Buck DC-DC Converter*

2.1 Basic Operation and Working Principle

The buck converter operates by rapidly switching a power semiconductor (e.g., MOSFET) on and off, controlling energy transfer to the load. Here's how it works:

Step 1: Switch ON State

- The MOSFET (switch) is turned ON by a PWM (Pulse Width Modulation) signal.
- Current flows from V_{in} through the inductor (L) to the load and output capacitor (C).
- The inductor stores energy in its magnetic field, and the capacitor charges.
- The diode (D) is reverse-biased (off) during this phase.

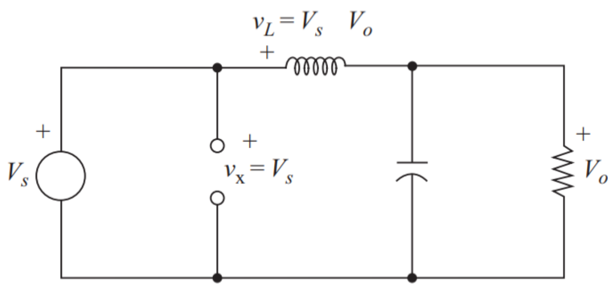

Figure 2: *Equivalent circuit for the switch closed*

Step 2: Switch OFF State

- The MOSFET is turned OFF.
- The inductor's magnetic field collapses, maintaining current flow through the free-wheeling diode (D).
- Energy stored in L and C continues to power the load.

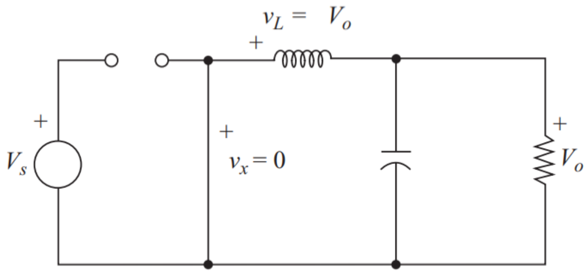

Figure 3: *Equivalent circuit for the switch open*

2.2 Key Components

A buck converter comprises five critical components:

- **MOSFET (Switch):**
  - Acts as a fast-switching device (e.g., N-channel MOSFET).
  - Controlled by a PWM signal to regulate energy transfer.
- **Diode (Freewheeling Diode):**
  - Provides a path for inductor current when the switch is OFF.
  - Modern designs often replace diodes with synchronous rectifiers (low-side MOSFETs) for higher efficiency.
- **Inductor (L):**
  - Stores energy during the ON state and releases it during the OFF state.
  - Smooths current ripple.
- **Capacitor (C):**
  - Filters output voltage ripple caused by switching.
  - Maintains a stable V_{out} during load transients.

2.3 Duty Cycle, In/Output Voltage Relationship, and Efficiency

1. **Duty Cycle (D)**
   - Defined as the ratio of ON time (T_{on}) to the total switching period (T_{sw}):

$$D = \frac{T_{\text{on}}}{T_{\text{sw}}}$$

2. **Voltage Relationship:**
   - In continuous conduction mode (CCM), the output voltage is:
   $$V_{\text{out}} = D \cdot V_{\text{in}}$$
   - This assumes ideal components (no losses).
3. **Efficiency** (η)
   - Efficiency is the ratio of output power (P_{out}) to input power (P_{in}):
   $$\eta = \frac{P_{\text{out}}}{P_{\text{in}}} \times 100\%$$
4. **Losses reduce efficiency:**
   - **Conduction Losses:** I^2R losses in MOSFET, diode, inductor, and PCB traces.
   - **Switching Losses:** Energy lost during MOSFET turn-on/turn-off transitions.
   - **Inductor Core Losses:** Hysteresis and eddy current losses in the inductor.
   - **Capacitor ESR Losses:** Equivalent series resistance (ESR) in the capacitor.

3 Mathematical Modeling

This section explains the mathematical foundation of buck converters, including voltage conversion equations, conduction modes, and ripple calculations.

3.1 Derivation of the Voltage Conversion Equation

During the ON state:

- The switch is closed, and the inductor is charged by the input voltage.
- The voltage across the inductor is:
$$V_L = V_{\text{in}} - V_{\text{out}}$$
- The current through the inductor increases linearly according to:
$$\Delta I_L = \frac{V_L \cdot T_{\text{on}}}{L} = \frac{(V_{\text{in}} - V_{\text{out}}) \cdot T_{\text{on}}}{L}$$

During the OFF state:

- The switch is open, and the inductor discharges energy to the load through the diode.
- The voltage across the inductor is:
$$V_L = -V_{\text{out}}$$
- The current through the inductor decreases linearly according to:
$$\Delta I_L = \frac{V_L \cdot T_{\text{off}}}{L} = \frac{-V_{\text{out}} \cdot T_{\text{off}}}{L}$$
- In steady-state operation, the increase in inductor current during the ON state equals the decrease during the OFF state:
$$L \cdot (V_{\text{in}} - V_{\text{out}}) \cdot T_{\text{on}} = L \cdot V_{\text{out}} \cdot T_{\text{off}}$$
- Simplifying this equation gives:
$$V_{\text{out}} = D \cdot V_{\text{in}}$$

3.2 Continuous Conduction Mode (CCM)vsDiscontinuous Conduction Mode (DCM)

Continuous Conduction Mode

- In **Continuous Conduction Mode (CCM)**, the inductor current never drops to zero during the switching cycle.
- This mode is common in high-power applications where efficiency and stability are critical.

Discontinuous Conduction Mode

- In **Discontinuous Conduction Mode (DCM)**, the inductor current drops to zero before the next switching cycle begins.
- This mode is often used in low-power applications where simplicity and cost are more important than efficiency.

Key Differences Between CCM and DCM

Inductor Current Waveform:

- In **CCM**, the inductor current is continuous and has a triangular waveform.
- In **DCM**, the inductor current becomes zero for part of the switching cycle.

Output Voltage Relationship:

- In **CCM**, the output voltage is determined solely by the duty cycle:

$$V_{\text{out}} = D \cdot V_{\text{in}}$$

- In **DCM**, the output voltage depends on the load current and switching frequency:

$$V_{\text{out}} = \frac{2 \cdot L \cdot f_{\text{sw}} \cdot R_{\text{load}}}{V_{\text{in}} \cdot D^2}$$

Efficiency:

- **CCM** is generally more efficient because the inductor is fully utilized.
- **DCM** can suffer from higher losses due to the discontinuous nature of the current.

Boundary Condition Between CCM and DCM:

- The boundary between **CCM** and **DCM** occurs when the inductor current just reaches zero at the end of the OFF state.
- The critical inductance L_{crit} for maintaining CCM is given by:

$$L_{\text{crit}} = \frac{(1 - D) \cdot R_{\text{load}}}{2 \cdot f_{\text{sw}}}$$

- If the actual inductance L is greater than L_{crit}, the converter operates in **CCM**. Otherwise, it operates in **DCM**.

Ripple Current in the Inductor:

- The ripple current ΔI_L is the peak-to-peak variation in the inductor current during a switching cycle.
- For **CCM**, the ripple current is given by:

$$\Delta I_L = \frac{V_{\text{in}} \cdot D \cdot (1 - D)}{L \cdot f_{\text{sw}}}$$

  where:

  - V_{in} is the input voltage.
  - D is the duty cycle.
  - L is the inductance.
  - f_{sw} is the switching frequency.

Ripple Voltage Across the Capacitor:

- The ripple voltage ΔV_C is the variation in the output voltage caused by the charging and discharging of the capacitor.
- It is given by:

$$\Delta V_C = \frac{\Delta I_L}{8 \cdot f_{\text{sw}} \cdot C}$$

  where:

  - ΔI_L is the ripple current.
  - C is the capacitance.

Operation Waveform of CCM

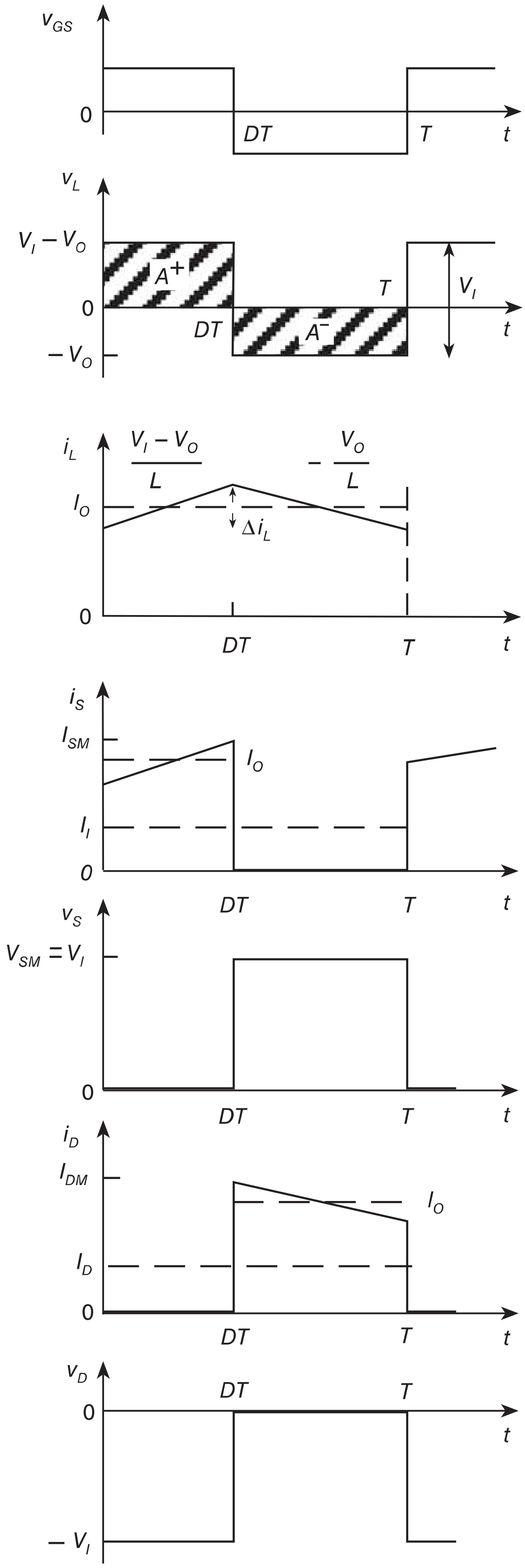

Figure 4: *Idealized current and voltage waveforms in the PWM buck converter for CCM.*

Operation Waveform of DCM

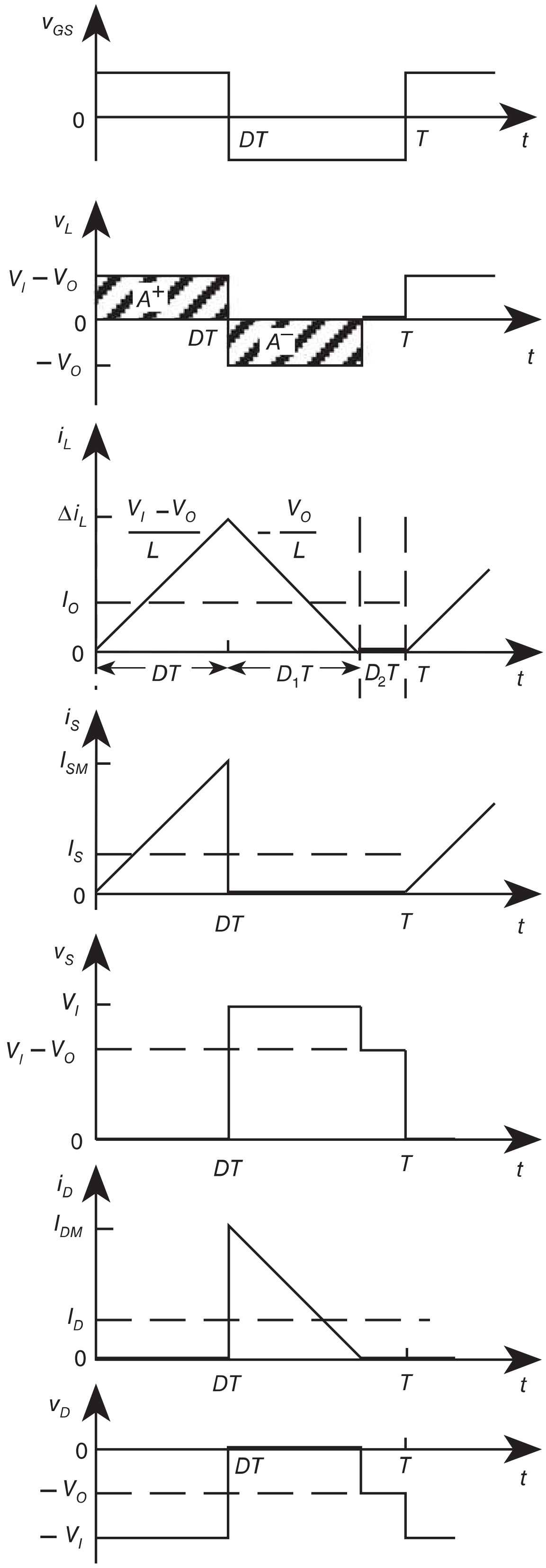

Figure 5: *Idealized current and voltage waveforms in the PWM buck converter for DCM.*

4 Design and Implementation

4.1 Power Stage Calculation and Component Selection

Design of a 120W Buck Converter Operating in Continuous Conduction Mode (CCM) with the following specifications:

Parameter	**Value**
Output Voltage (V_{out})	24 V
Input Voltage (V_{in})	30 V
Output Power (P_{out})	120 W
Switching Frequency (f_{sw})	100 kHz
Ripple Current Rated	30% of I_{out}
Ripple Voltage Rated	1% of V_{out}

Table 1: Parameters Needed for Power Stage Calculation

Power Stage Calculations

1. Duty Cycle (D)

$$D = \frac{V_{out}}{V_{in}} = \frac{24}{30} = \boxed{0.8\,(80\%)}$$

2. Output Current (I_{out})

$$I_{out} = \frac{P_{out}}{V_{out}} = \frac{120}{24} \approx \boxed{5\,\text{A}}$$

3. Output Resistance (R)

$$R = \frac{V_{out}}{I_{out}} = \frac{24}{5} \approx \boxed{4.8\,\Omega}$$

4. Inductor Ripple Current (ΔI_L)

$$\Delta I_L = 30\% \times I_{out} = 0.3 \times 5 \approx \boxed{1.5\,\text{A (peak-to-peak)}}$$

5. Maximum/Minimum Inductor Current

$$I_{L_{\max}} = I_{out} + \frac{\Delta I_L}{2} = 5 + 0.75 \approx \boxed{5.75\,\text{A}}$$

$$I_{L_{\min}} = I_{out} - \frac{\Delta I_L}{2} = 5 - 0.75 \approx \boxed{4.25\,\text{A}}$$

6. Ripple Voltage (ΔV_{out})

$$\Delta V_{out} = 1\% \times V_{out} = 0.01 \times 24 = \boxed{0.24\,\text{V (peak-to-peak)}}$$

7. Output Capacitor (C)

$$C = \frac{\Delta I_L \cdot D}{8 \cdot F_s \cdot \Delta V_{out}} = \frac{1.5 \cdot 0.8}{8 \cdot 100\,\text{kHz} \cdot 0.24} = \boxed{6.25\,\mu\text{F}}$$

8. Inductor (L)

$$L = \frac{V_{out} \cdot (V_{in} - V_{out})}{V_{in} \cdot F_s \cdot \Delta I_L} = \frac{24 \cdot (30 - 24)}{30 \cdot 100\,\text{kHz} \cdot 1.5} = \boxed{32\,\mu\text{H}}$$

9. MOSFET Selection

Table 2: IRF540N MOSFET

Parameter	Value	Test Conditions
V_{DS}	100 V	Absolute Maximum Rating
I_D	14 A	$T_C = 25\,°\text{C}$
	10 A	$T_C = 100\,°\text{C}$
$R_{DS(on)}$	0.16 Ω	$V_{GS} = 10\,\text{V}, I_D = 8.4\,\text{A}$

10. Diode Selection

Table 3: 15SQ045 Schottky Diode

Parameter	Value	Test Conditions
V_{RRM}	45 V	Peak Reverse Voltage
I_F	15 A	Continuous, $T_A = 25\,°\text{C}$

11. Inductor Design

•Inductor Core Selection Using Kg Method

Table 4: Input Parameters for Ferrite Core Design

Symbol	Value	Description
L	32 µH	Inductance
I_o	5 A	DC Current
ΔI	1.5 A	AC Ripple Current
P_o	120 W	Output Power
α	1%	Regulation
f_s	100 kHz	Switching Frequency
B_m	0.25 T	Operating Flux Density
K_u	0.4	Window Utilization
T_r	25 °C	Temperature Rise Goal

- **Peak Current**

$$I_{\text{pk}} = I_o + \frac{\Delta I}{2} = 5 + \frac{1.5}{2} = \boxed{5.75\,\text{A}}$$

- **Energy-Handling Capability**

$$E = \frac{1}{2} L I_{\text{pk}}^2 = \frac{1}{2} \cdot 32 \times 10^{-6} \cdot (5.75)^2 = \boxed{5.29 \times 10^{-4}\,\text{J}}$$

- **Electrical Condition Coefficient**

$$K_e = 0.145 \cdot P_o \cdot B_m^2 \cdot 10^{-4} = 0.145 \cdot 120 \cdot (0.25)^2 \cdot 10^{-4} = \boxed{1.08 \times 10^{-4}}$$

- **Core Geometry Coefficient**

$$K_g = \frac{E^2}{K_e \cdot \alpha} = \frac{(4.87 \times 10^{-4})^2}{9.96 \times 10^{-5} \cdot 1} = \boxed{2.6 \times 10^{-3}\,\text{cm}^5}$$

Core type (*A*) (mm)	Geometrical constant K_g cm^5	Geometrical constant K_{gfe} cm^x	Cross-sectional area A_c (cm^2)	Bobbin winding area W_A (cm^2)	Mean length per turn *MLT* (cm)	Magnetic path length l_m (cm)	Core weight (g)
EE12	$0.731 \cdot 10^{-3}$	$0.458 \cdot 10^{-3}$	0.14	0.085	2.28	2.7	2.34
EE16	$2.02 \cdot 10^{-3}$	$0.842 \cdot 10^{-3}$	0.19	0.190	3.40	3.45	3.29
EE19	$4.07 \cdot 10^{-3}$	$1.3 \cdot 10^{-3}$	0.23	0.284	3.69	3.94	4.83
EE22	$8.26 \cdot 10^{-3}$	$1.8 \cdot 10^{-3}$	0.41	0.196	3.99	3.96	8.81

Figure 6: *Pulse and Magnetizing Inductance Current Waveform*

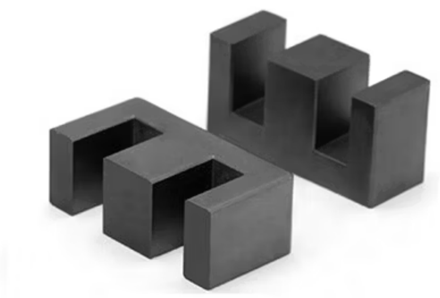

Figure 7: *EE Core Selected*

Parameter	**Value**	**Description**
MPL	3.94	Magnetic Path Length (cm)
We	4.83	Core Weight (g)
MLT	3.69	Mean Length per Turn (cm)
A_c	0.23	Cross-Sectional Area (cm^2)
W_a	0.284	Window Area (cm^2)
A_p	0.0653	Core Product (cm^4)
K_g	4.07×10^{-3}	Core Geometry (cm^5)

Table 5: Core Parameters for Ferrite Design

- **Current Density (J)**

$$J = \frac{2 \cdot E \cdot 10^4}{B_m \cdot A_p \cdot K_u} = \frac{2 \cdot 5.29 \times 10^{-4} \cdot 10^4}{0.25 \cdot 0.06532 \cdot 0.4} \approx 1620\,\mathrm{A\,cm^{-2}}.$$

- **RMS Current ($I_{\mathbf{RMS}}$)**

$$I_{\text{rms}} = \sqrt{I_o^2 + \Delta I^2} = \sqrt{5^2 + 1.5^2} = \boxed{5.22\,\mathrm{A}}.$$

- **Bare Wire Area (A_w)**

$$A_w = \frac{I_{\text{rms}}}{J} = \frac{5.22}{1620} = \boxed{0.003\,22\,\mathrm{cm}^2}.$$

- **Selected Bare Wire Area**

$$A_{ws} = 0.01\,\mathrm{cm}^2$$

- **Effective Window Area ($W_{\mathbf{aeff}}$)**
  - $S_3 = 0.75$ Window Effective factor for Ferrite core

$$W_{\text{aeff}} = W_a \cdot S_3 = 0.284 \cdot 0.75 = \boxed{0.213\,\mathrm{cm}^2}.$$

- **Number of Turns (N)**
  - $S_2 = 0.6$ Fill Factor

$$N = \frac{W_{\text{aeff}} \cdot S_2}{A_{ws}} = \frac{0.213 \cdot 0.6}{0.01} = \boxed{12.7\ \text{turns}}.$$

- **Required Air Gap (L_g)**

$$L_g = \frac{0.4\pi N^2 A_c \cdot 10^{-8}}{L} - \frac{MPL}{\mu_0}$$

Substituting values:

$$L_g = \frac{0.4 \cdot \pi \cdot 10^2 \cdot 0.23 \cdot 10^{-8}}{32 \times 10^{-6}} - \frac{3.94}{2500} = \boxed{0.0137\,\mathrm{cm}}.$$

4.2 Power Stage Calculation and Component Selection Using Matlab Script

Below is the MATLAB code use to calculate the Power Stage and Component Selection for the 120W Buck Converter design.

```
%% 120W Buck Converter Design
clc, clear, close all;
%% Design Parameter
Vin = 30;                 % Input Voltage (V)
Vout = 24;                % Output Voltage (V)
Pout = 120;               % Output Power (W)
Fs = 100e3;               % Switching Frequency (Hz)
ripple_current_rate = 0.3; % 30% ripple current
ripple_voltage_rate = 0.01; % 1% ripple voltage
%% 1. Duty Cycle (D)
D = Vout / Vin;
%% 2. Output Current (Iout)
Iout = Pout / Vout;
%% 3. Output Resistance (Rout)
Rout = Vout / Iout;
%% 4. Ripple Current
delta_I_L = ripple_current_rate * Iout;
I_L_max = Iout + delta_I_L / 2;
I_L_min = Iout - delta_I_L / 2;
%% 5. Inductor (L)
L = (Vout * (Vin - Vout)) / (Vin * Fs * delta_I_L);
L_standard = ceil(L / 1e-6); % Round to nearest standard valuellt
%% 6. Ripple Voltage
delta_Vout = ripple_voltage_rate * Vout;
%% 7. Output Capacitor (C)
C = (delta_I_L * D) / (8 * Fs * delta_Vout);
C_standard = ceil(C / 1e-6); % Round to nearest standard value
```

Listing 1: MATLAB Code for 120W Buck Converter Design

4.3 Simulation On LTspice (Open Loop)

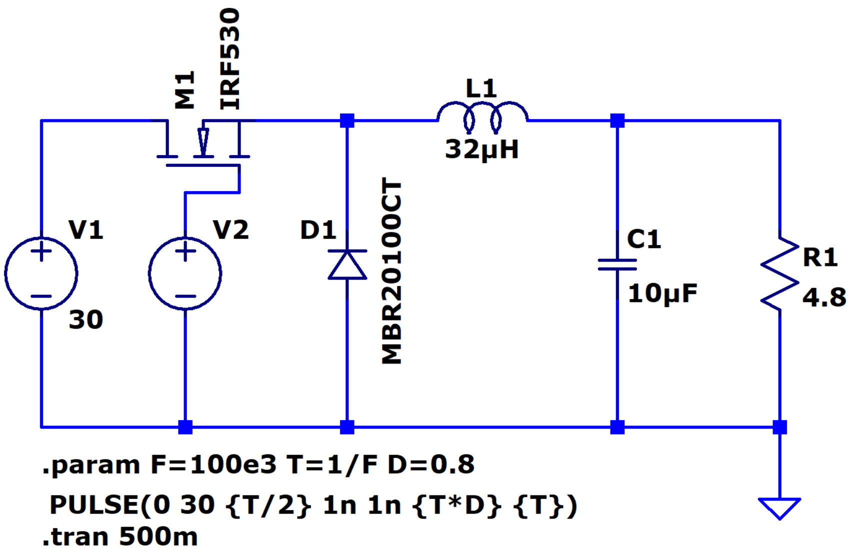

Figure 8: *Buck Converter Schematic on LTspice*

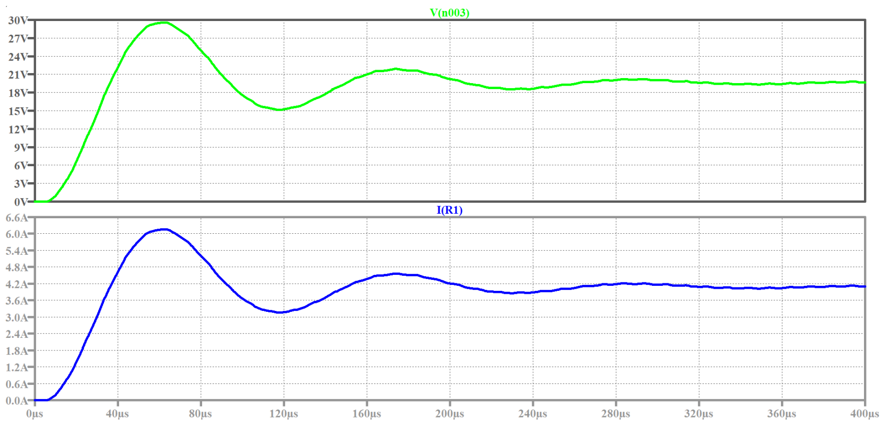

Figure 9: *Output Voltage and Ouput Current Waveform*

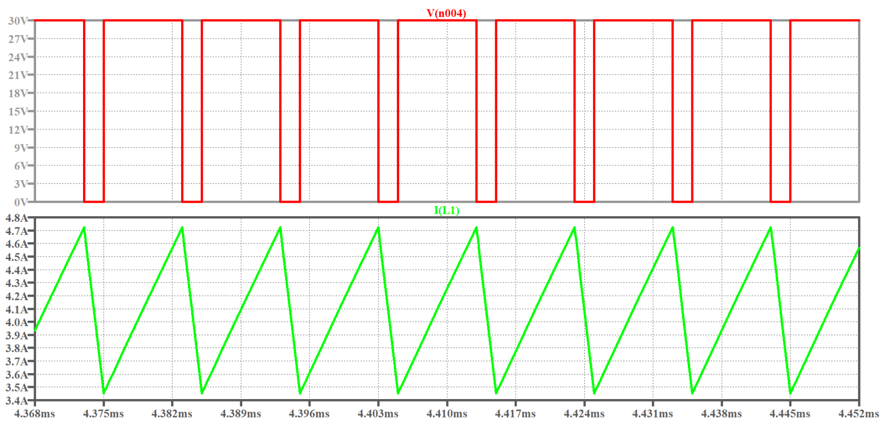

Figure 10: *Duty Cycle and Inductor Current Waveform*

4.4 Simulation On Simulink (Open Loop)

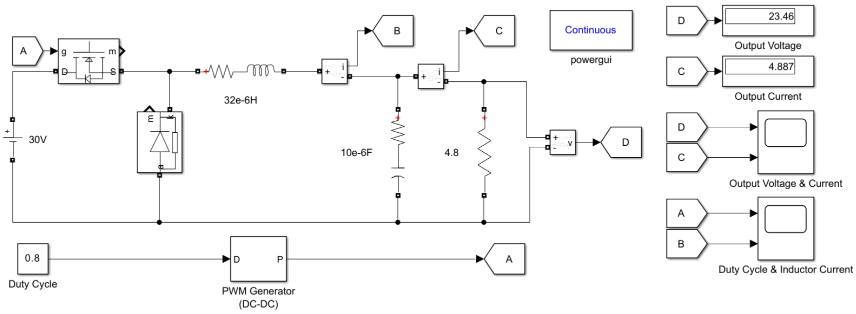

Figure 11: *Buck Converter Schematic on LTspice*

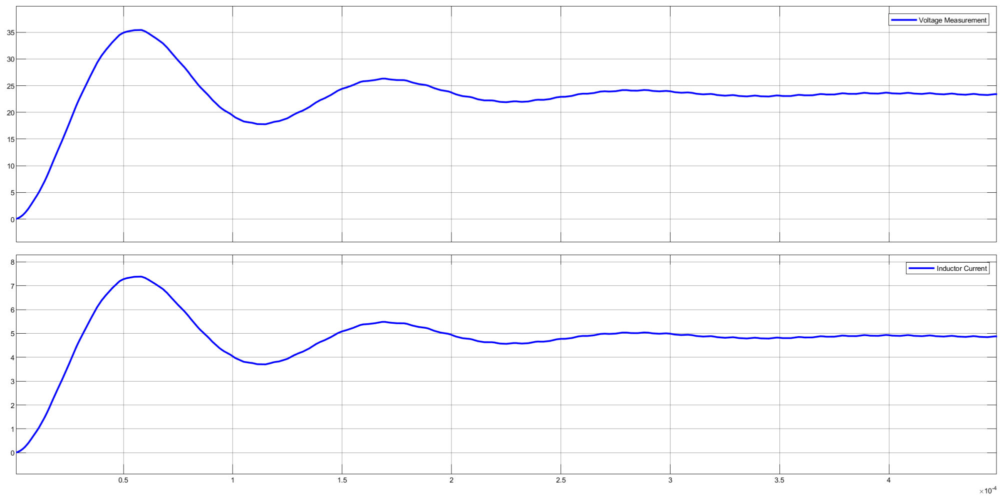

Figure 12: *Output Voltage and Ouput Current Waveform*

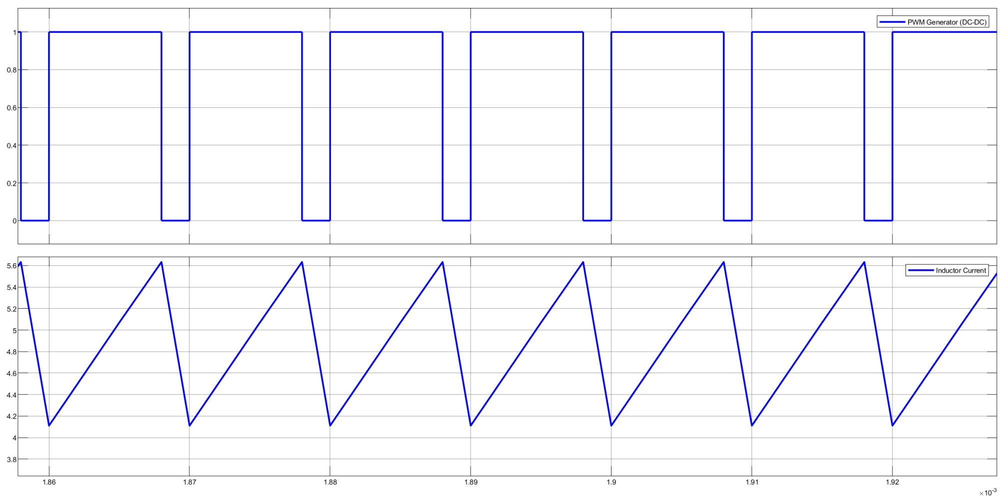

Figure 13: *Duty Cycle and Inductor Current Waveform*

4.5 Simulation on Simulink Using PI Controller (Close Loop)

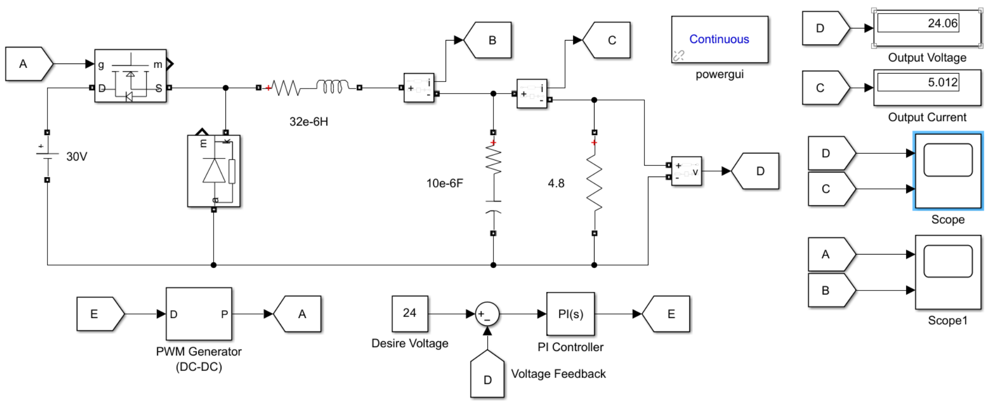

Figure 14: *Buck Converter Schematic on LTspice*

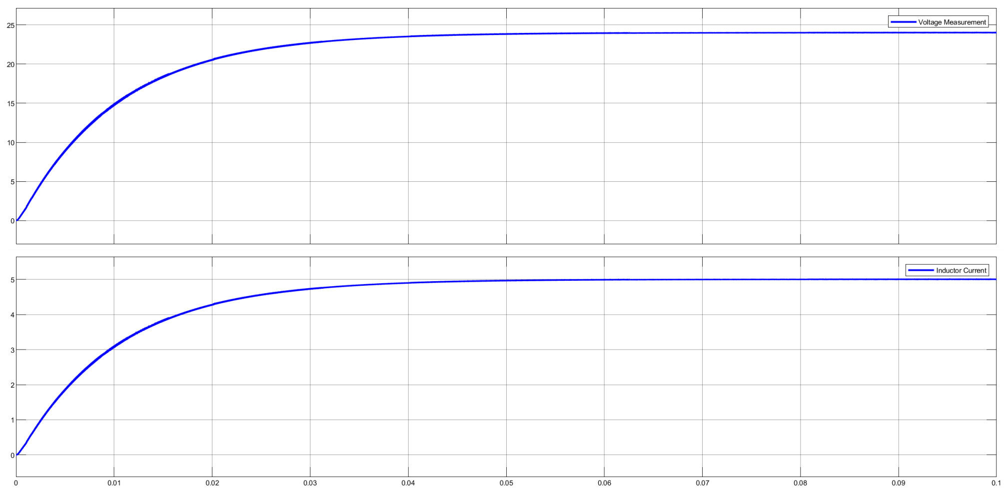

Figure 15: *Output Voltage and Ouput Current Waveform*

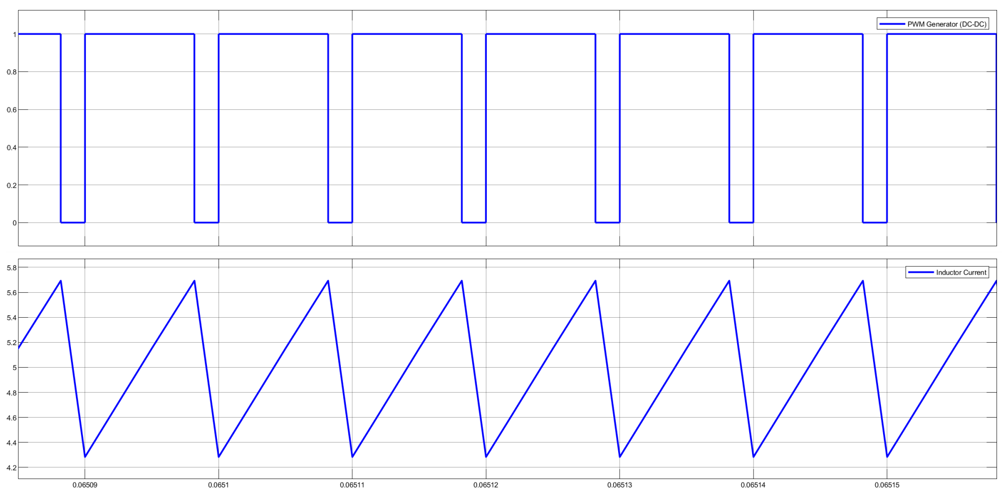

Figure 16: *Duty Cycle and Inductor Current Waveform*

References

[1] COLONEY WM.T. MCLYMAN, *Transformer Design and Inductor Design Handbook*, 3rd ed., Publisher Marcel Dekker, Inc, California, U.S.A, 2004, ISBN: 0-8247-5393-3.